This week's task is to gain "the empirical understanding of what is going on". I will work on coding my own SAE in a way I can easily tweak and visualize the effects of different parameters. In other words, I will be learning PyTorch - a deep learning framework commonly perceived as more appropriate for research compared to Tensorflow.

I think an important distinction of the common use case of SAE to the one here should be made. Usually, SAE are used to disentangle dense representations (e.g. word embeddings) into more interpretable ones, while here the goal is to learn disentangled and interpretable features directly from the data (MNIST handwritten numbers).

Initial attempt and tests

To find out how to implement SAE I looked over some tutorials and simply searched for implementations and started with 2 versions:

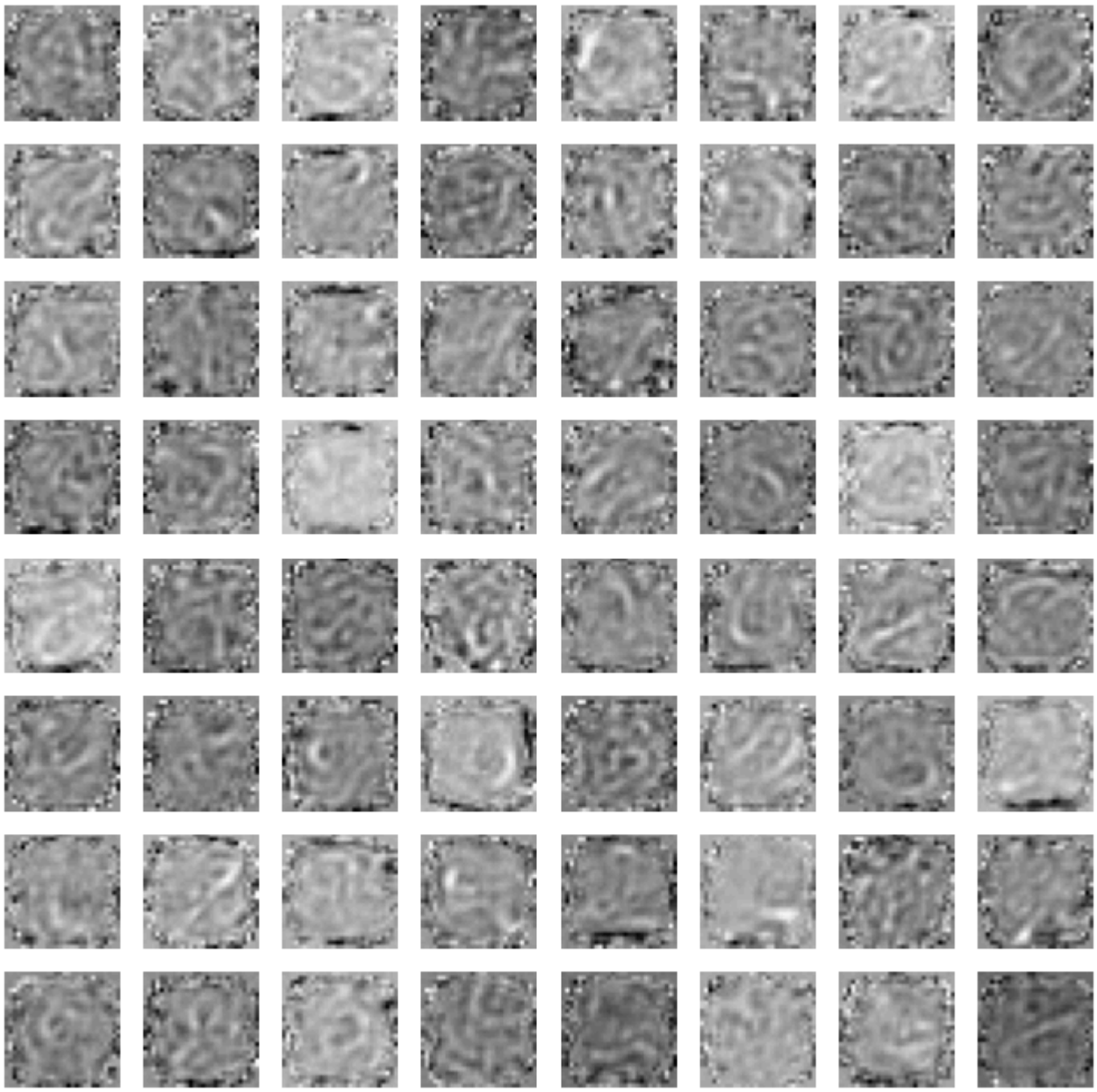
- Geeks for Geeks - Tensorflow - https://www.geeksforgeeks.org/deep-learning/sparse-autoencoders-in-deep-learning/
- ChatGPT implementation - Pytorch (you can find in my gh)

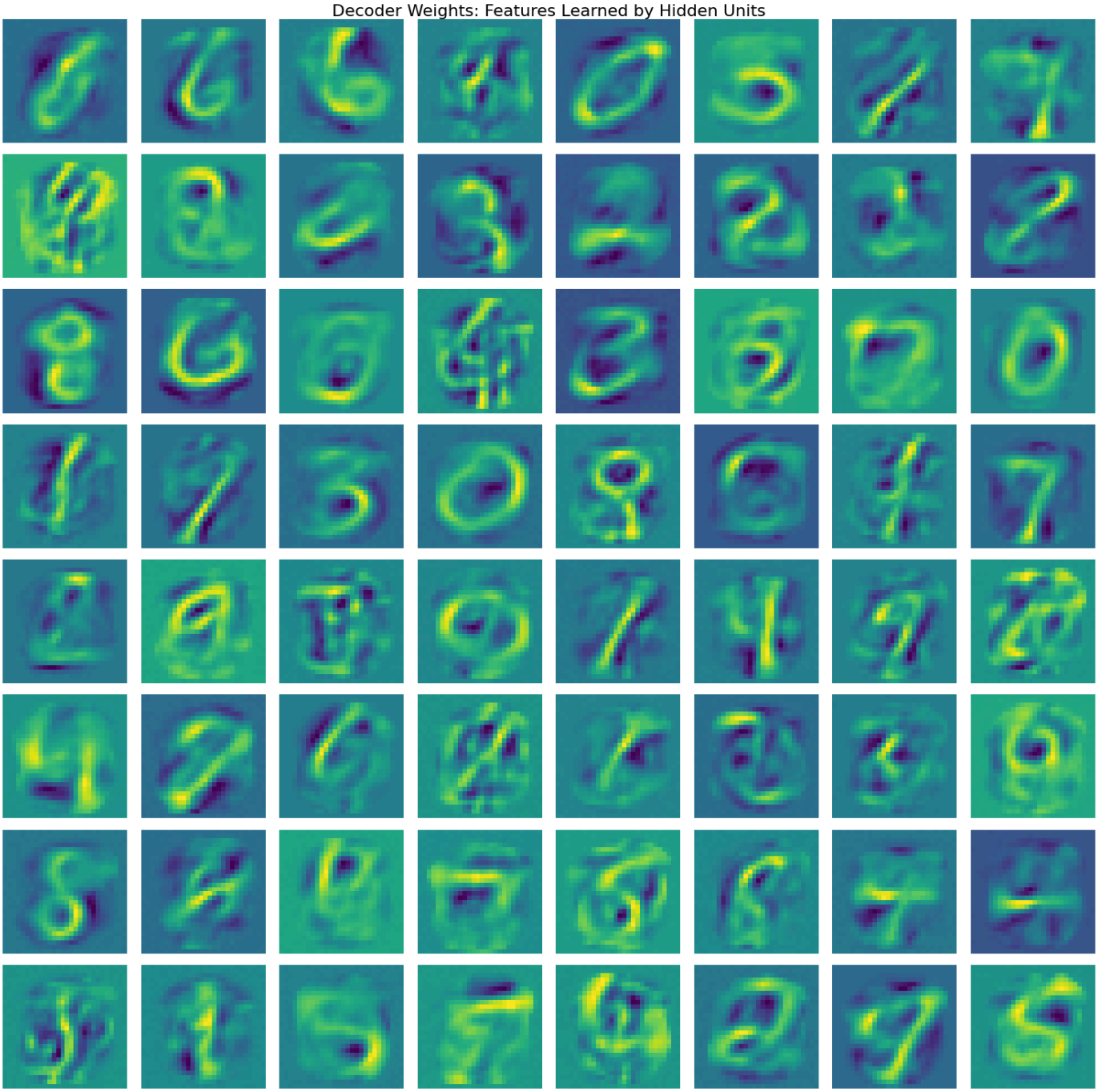
What I immediately noticed was the loss calculation.

```
def sparse_loss(y_true, y_pred):
    mse_loss = tf.reduce_mean(keras.losses.MeanSquaredError()(y_true, y_pred))
    hidden_layer_output = encoder(y_true)
    mean_activation = tf.reduce_mean(hidden_layer_output, axis=0)

    kl_divergence = tf.reduce_sum(sparsity_level * tf.math.log(sparsity_level / (mean_activation + 1e-
                                  (1 - sparsity_level) * tf.math.log((1 - sparsity_level) / (1 - mean_

    return mse_loss + lambda_sparse * kl_divergence
```

For the sparsity they use the kl_divergence Kullback–Leibler divergence $D_{KL} = \sum \left(\rho \log \frac{\rho}{\hat{\rho}+\epsilon} + (1-\rho) \log \frac{1-\rho}{1-\hat{\rho}+\epsilon} \right)$ wikipedia. When put in the loss function it instead of forcing sparsity (small number of non-zero activations) it rather penalizes the model for not having the average activation close to the desired sparsity level. This is a bit counter-intuitive, but seems to be common approach.

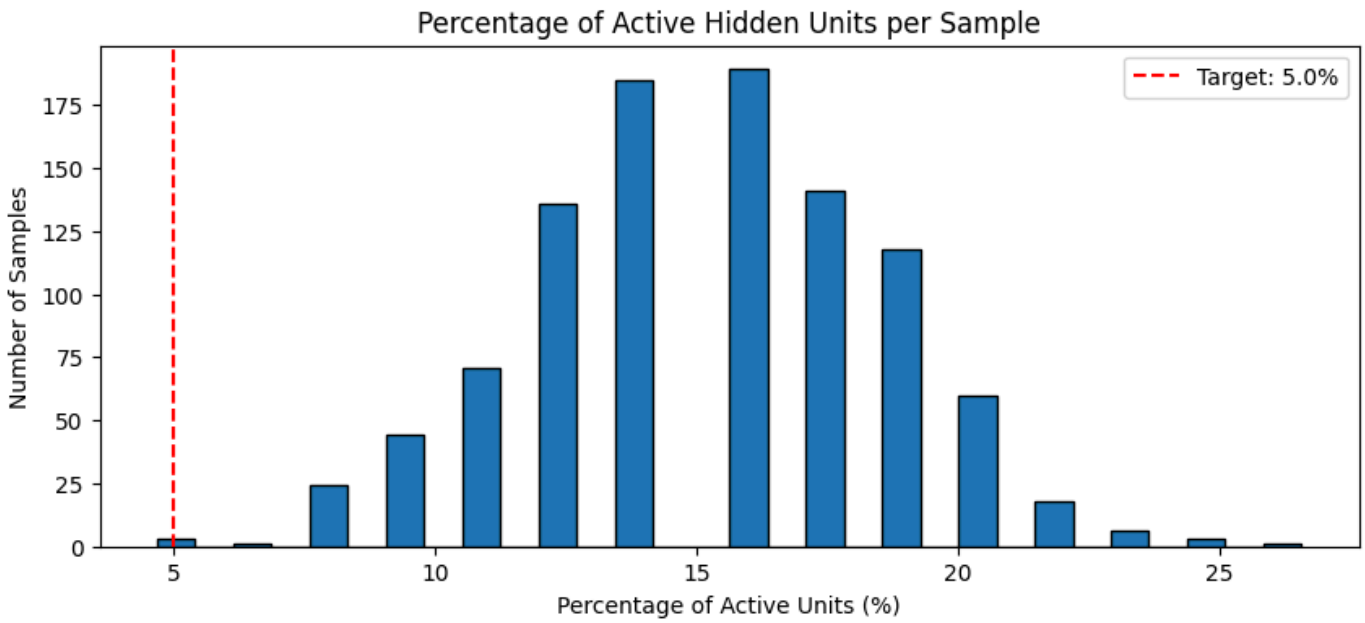
Visualization of the encoder/decoder weights after training:





Where the decoder weights do show some interpretable patterns, the encoder weights on the other hand are quite noisy.

But importantly, since the implementation forces the mean to be low, there are many active hidden units and the actual sparsity is pretty low:



My own implementation

As mentioned before, I decided to implement my own SAE in PyTorch. Deciding on the definition of the training objective I went to the paper from my last post - **Disentangling Dense Embeddings with Sparse Autoencoders - Charles O'Neill, Christine Ye, Kartheik Iyer, John F. Wu**, where they use $\boldsymbol{k}$-sparse constraint.

Conceptual Inconsistency in the Paper?

Quoting the paper:

> The training objective of our SAE combines three main components: a reconstruction loss, a sparsity constraint, and an auxiliary loss. The overall loss function is given by:
>
> $$\mathcal{L}(\theta, \phi) = \frac{1}{d}\|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \lambda \mathcal{L}_{\text{sparse}}(\mathbf{h}) + \alpha \mathcal{L}_{\text{aux}}(\mathbf{x}, \hat{\mathbf{x}})$$
>
> where $\lambda > 0$ and $\alpha > 0$ are hyperparameters controlling the trade-off between reconstruction fidelity, sparsity, and the auxiliary loss.
>
> For the sparsity constraint, we use a k-sparse constraint: only the k largest activations in $\mathbf{h}$ are retained, while the rest are set to zero (Makhzani et al., 2013; Gao et al., 2024). This approach avoids issues such as shrinkage, where L1 regularisation can cause feature activations to be systematically lower than their true values, potentially leading to suboptimal representations *shrinkage*, (Wright et al., 2024; Rajamanoharan et al., 2024).

I believe there is a conceptual inconsistency here. Using the $\boldsymbol{k}$-sparse constraint should not be in the loss function, as it is a hard constraint. Instead of penalizing activation of non k-largest activations, the smaller ones are zeroed out, thus the lambda variable seems useless.

> Note: After further investigation, the original paper this one cites indeeds clarifies this inconsistency like so:
>
> with $W_{\text{enc}} \in \mathbb{R}^{n \times d}$, $b_{\text{enc}} \in \mathbb{R}^n$, $W_{\text{dec}} \in \mathbb{R}^{d \times n}$, and $b_{\text{pre}} \in \mathbb{R}^d$. The training loss is defined by $\mathcal{L} = ||x - \hat{x}||_2^2 + \lambda ||z||_1$, where $||x - \hat{x}||_2^2$ is the reconstruction MSE, $||z||_1$ is an L_1 penalty promoting sparsity in latent activations z, and λ is a hyperparameter that needs to be tuned.
>
> **2.3 TopK activation function**
>
> We use a k-sparse autoencoder [Makhzani and Frey, 2013], which directly controls the number of active latents by using an activation function (TopK) that only keeps the k largest latents, zeroing the rest. The encoder is thus defined as:
>
> $$z = \text{TopK}(W_{\text{enc}}(x - b_{\text{pre}})) \tag{2}$$
>
> and the decoder is unchanged. The training loss is simply $\mathcal{L} = ||x - \hat{x}||_2^2$.
>
> where it correctly distinguishes between TopK and L1 sparse autoencoders.

Implementation

Moving forward, for now I disregard the auxiliary loss and implement the forward pass like so:

```
def forward(self, x):

    # Encoder: linear projection
    h_pre = self.encoder(x)

    # Apply ReLU to ensure non-negative activations
    h_pre = F.relu(h_pre)

    # TopK activation: keep only k largest activations per sample
    topk_vals, topk_idx = torch.topk(h_pre, self.k_param, dim=1)

    # Create sparse activation tensor
    h = torch.zeros_like(h_pre)
    h.scatter_(1, topk_idx, topk_vals)

    # Decoder: reconstruct input
    x_hat = self.decoder(h)

    return h, x_hat
```

https://docs.pytorch.org/docs/stable/generated/torch.nn.Module.html
https://docs.pytorch.org/docs/stable/generated/torch.nn.Sequential.html

and the loss calculation is a simple MSE.

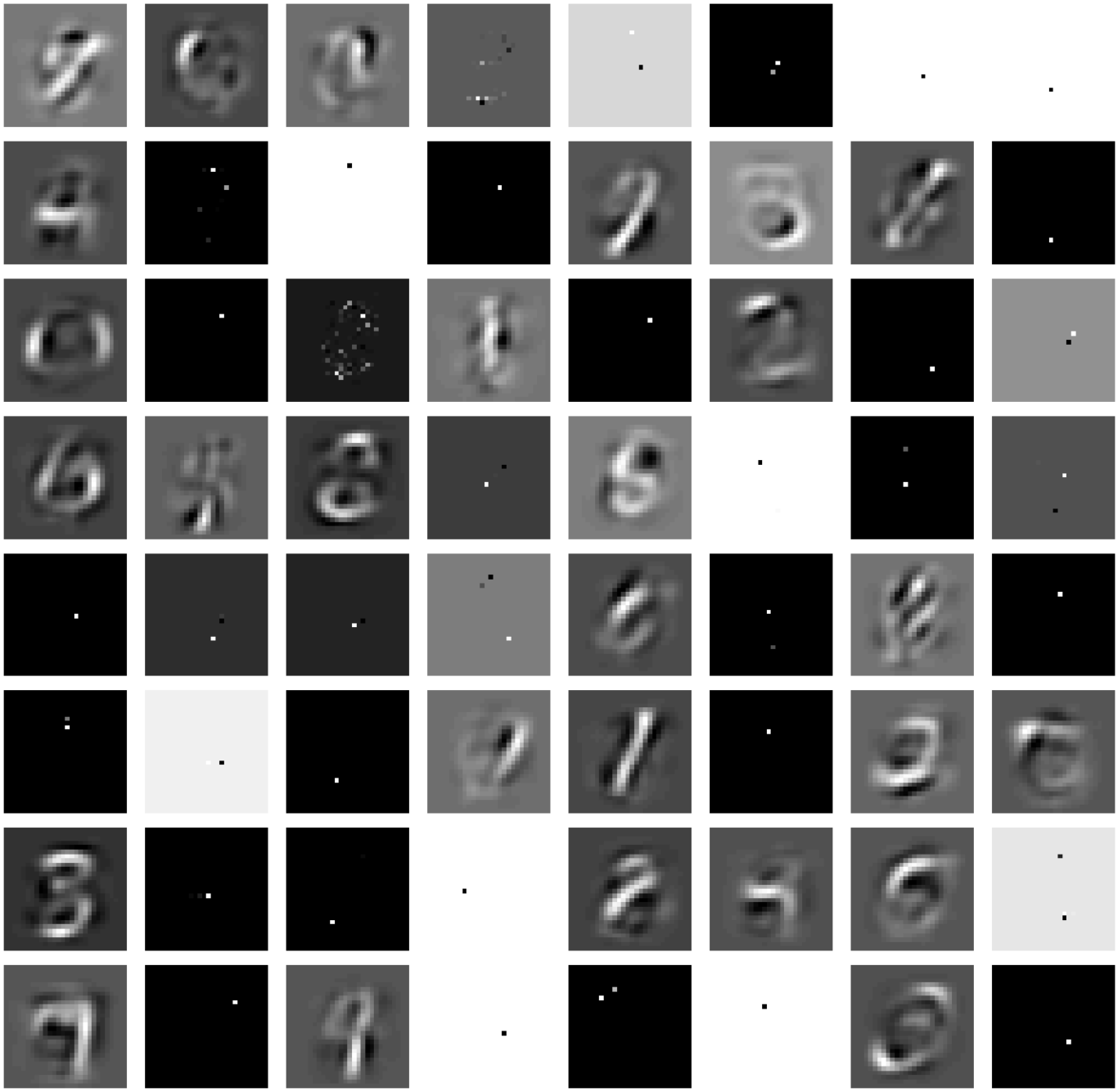
Results

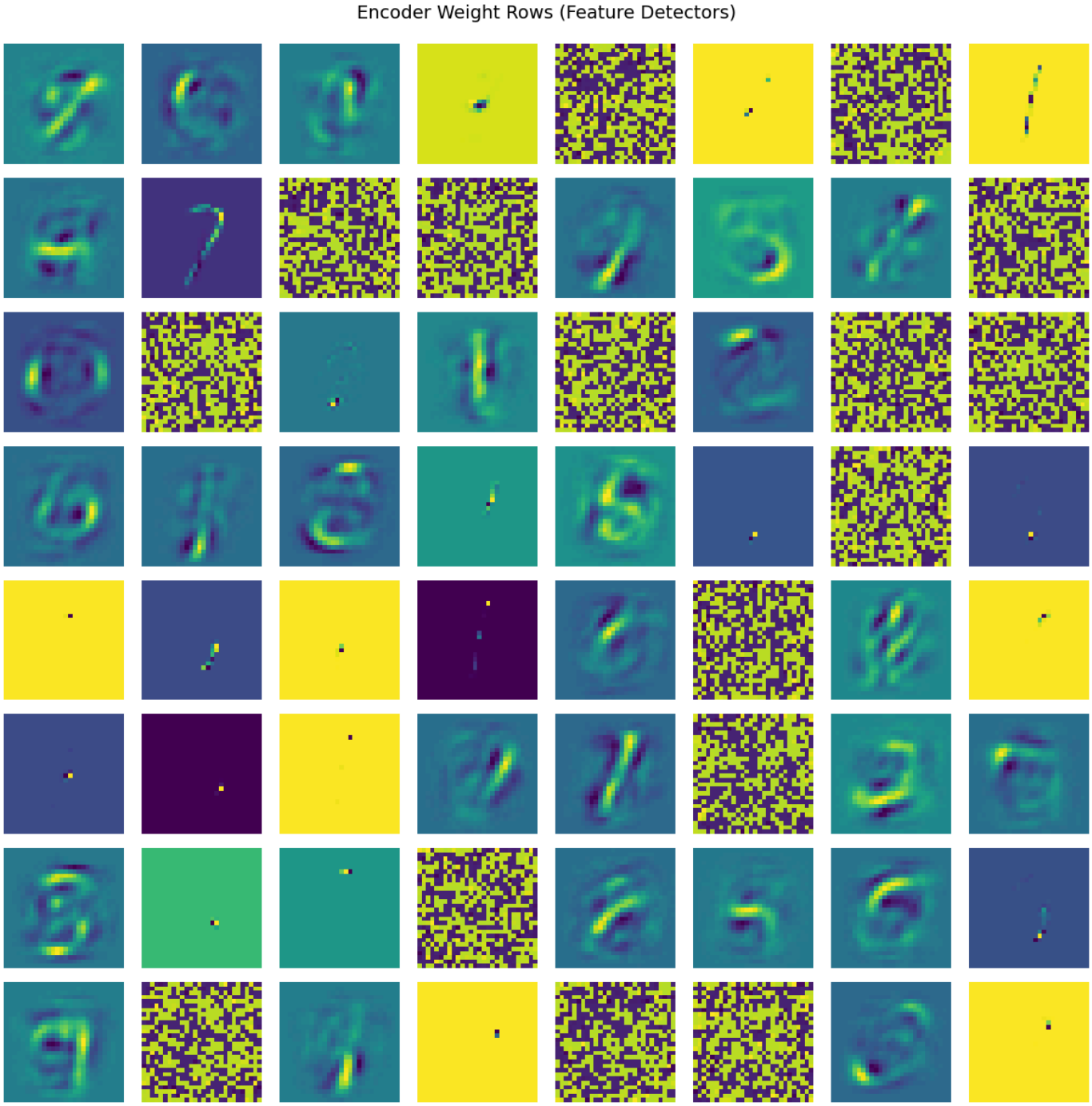
I decided to train with different values of k 10 and for comparison 64 features, while keeping other parameters constant:

```
learning_rate=0.0005,
num_epochs=50,
batch_size=256,
hidden_size=64,
```

k=10

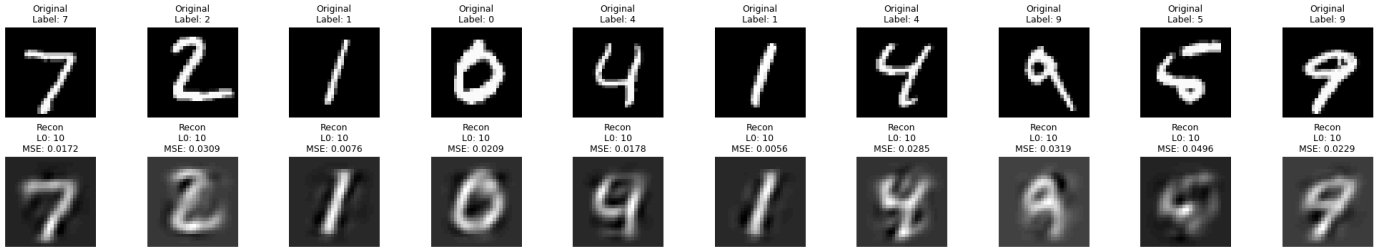
Decoder Weight Columns (Dictionary Atoms)

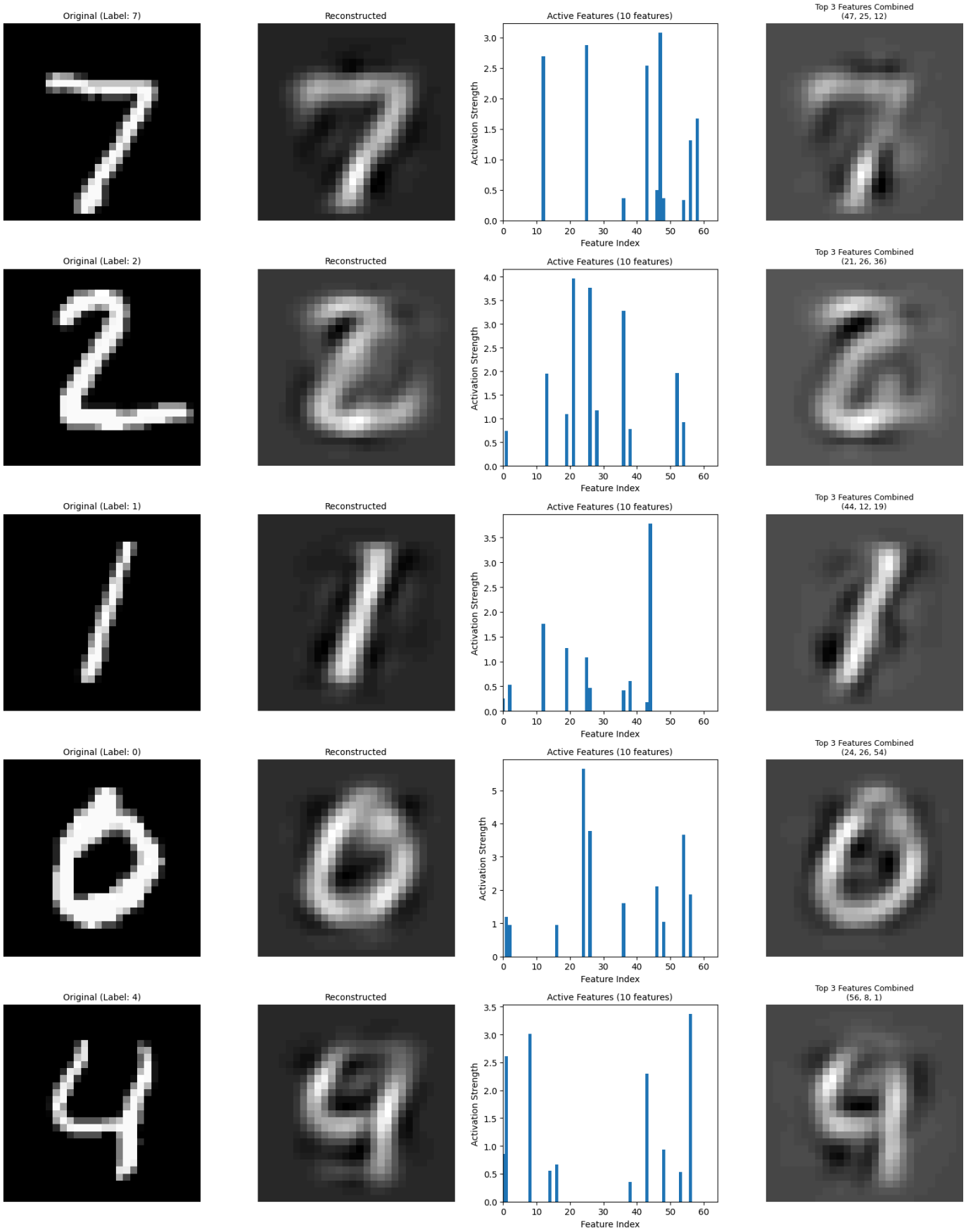




What shows quickly is the large number of dead features (>50%), but these encoder decoder weights show clear distinguishable patterns. Then each reconstruction is made with at most 10 features, which is quite impressive.

Original vs Reconstructed Images



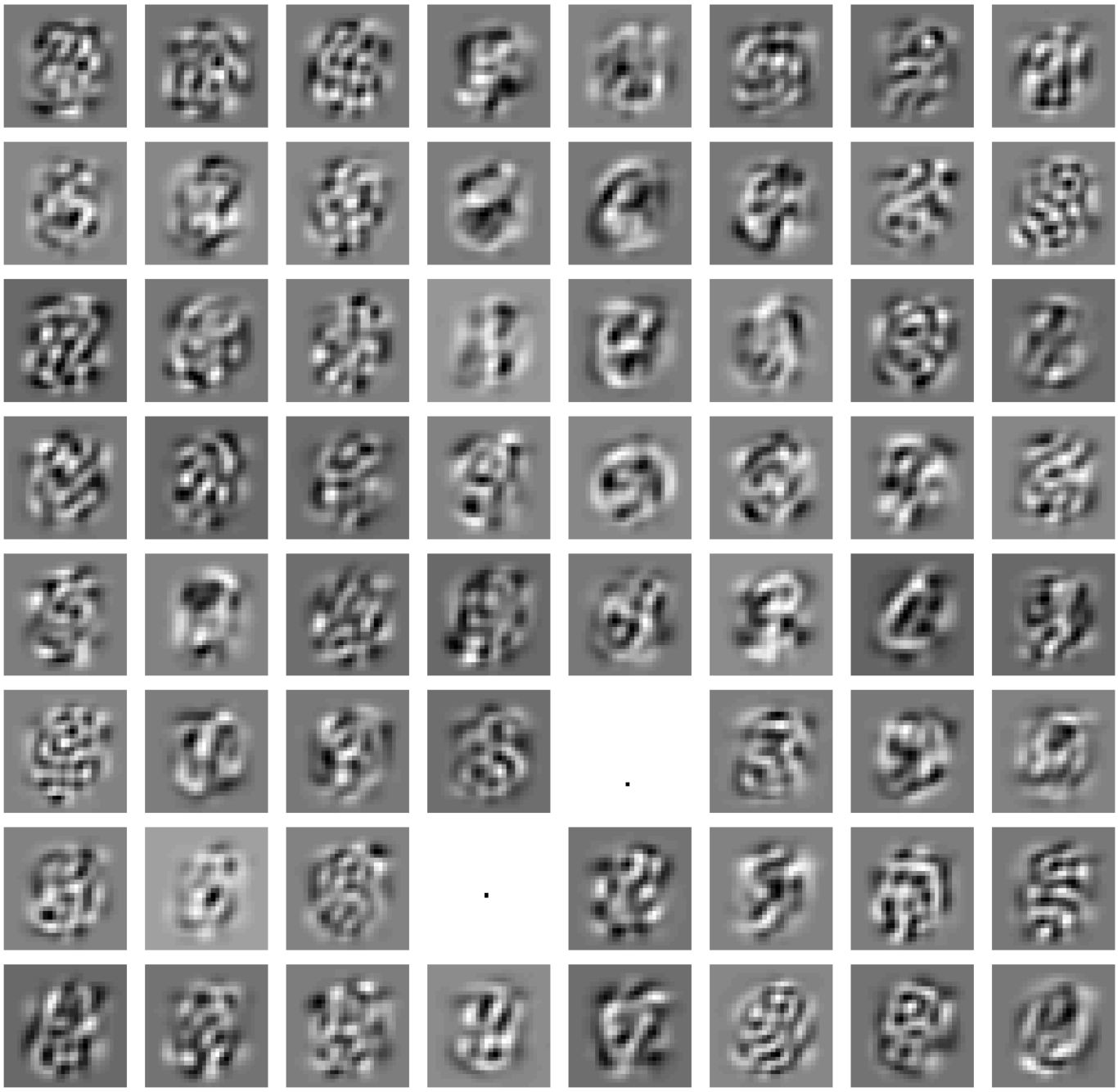


The main takeaway is that the model is not utilizing the full potential of available features, but the ones utilized are humanly interpretable.

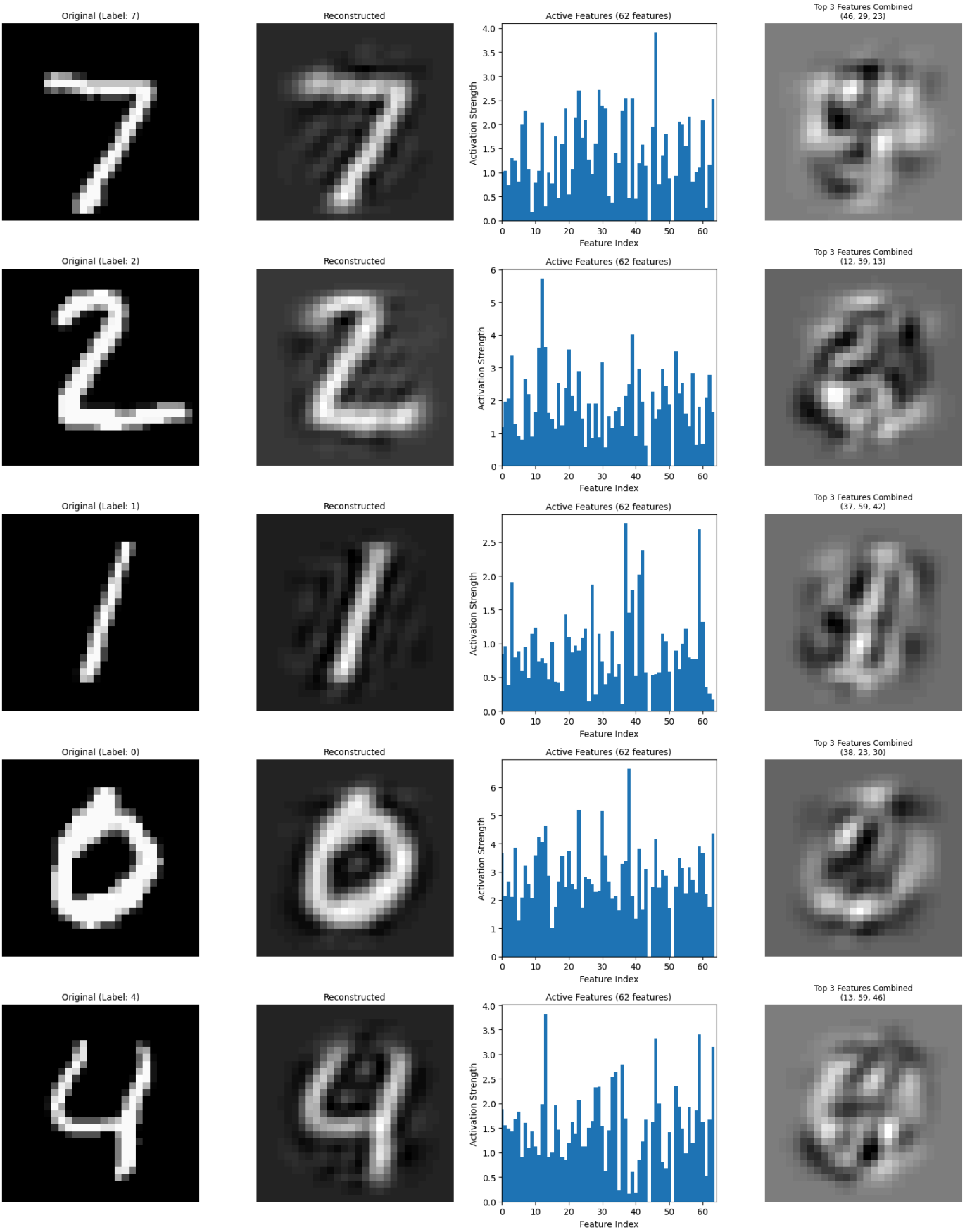
k=64

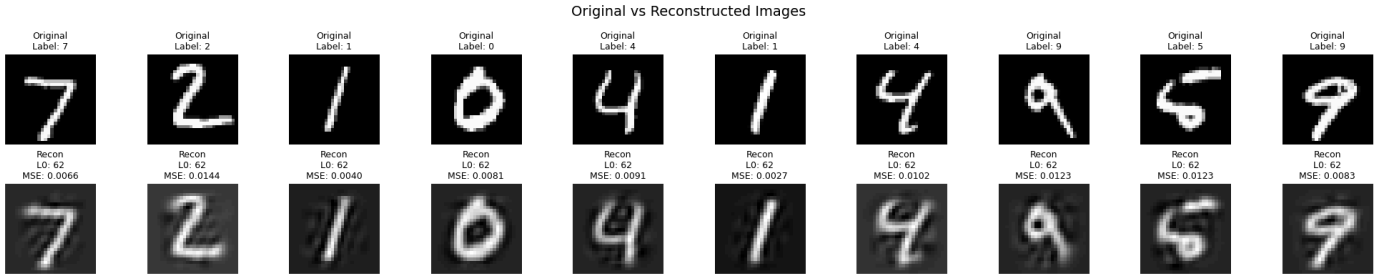
To compare, I trained the same model with k=64, which means no sparsity constraint at all. The results are quite different, for example the encoder weights look like so:

Decoder Weight Columns (Dictionary Atoms)



Which again contains dead features (2 in this case), but the rest of the features are not humanly interpretable.

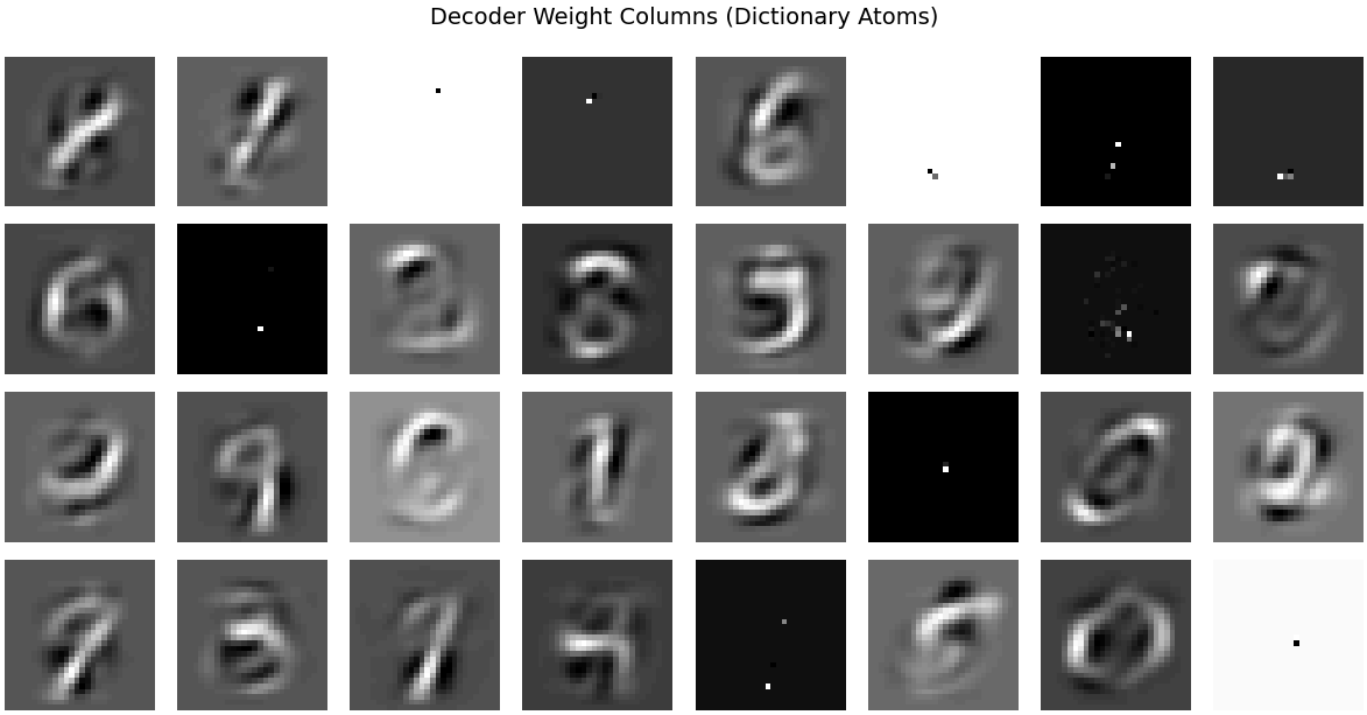

Original (Label: 7)
Reconstructed
Active Features (62 features)
Activation Strength
Feature Index
Top 3 Features Combined
(46, 29, 23)
Original (Label: 2)
Reconstructed
Active Features (62 features)
Activation Strength
Feature Index
Top 3 Features Combined
(12, 39, 13)
Original (Label: 1)
Reconstructed
Active Features (62 features)
Activation Strength
Feature Index
Top 3 Features Combined
(37, 59, 42)
Original (Label: 0)
Reconstructed
Active Features (62 features)
Activation Strength
Feature Index
Top 3 Features Combined
(38, 23, 30)
Original (Label: 4)
Reconstructed
Active Features (62 features)
Activation Strength
Feature Index
Top 3 Features Combined
(13, 59, 46)

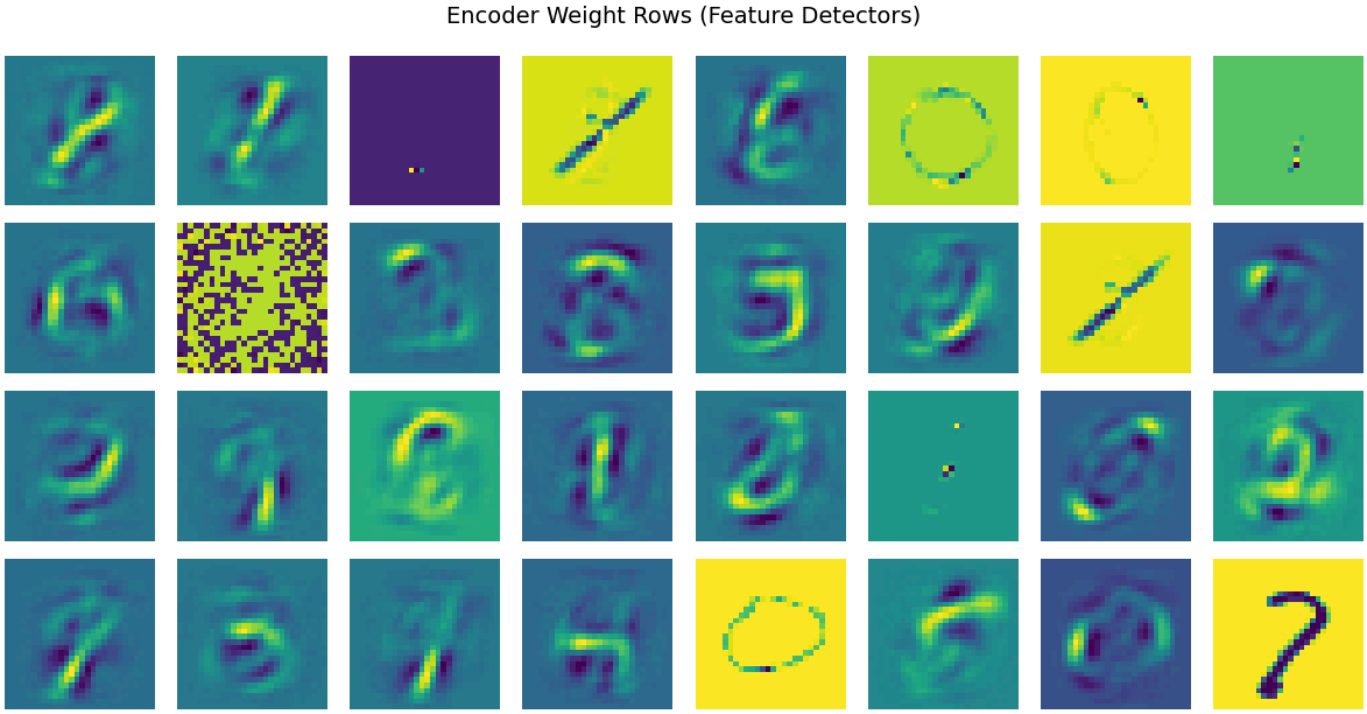


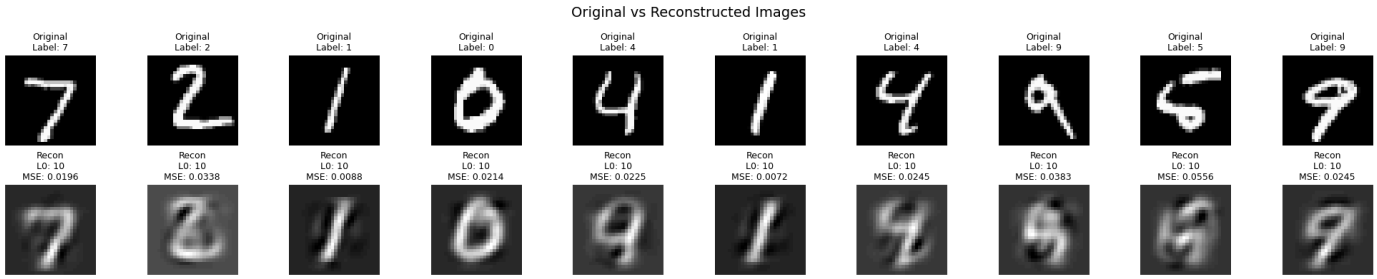
On the other hand, what one might have expected, when the model is less constrained, the reconstructions are better.

Smaller latent space?

With the large amount of dead features, I wondered if the model would perform similarly given that the latent space is smaller. Repeating the experiment with k=10, about 50% of features were dead, thus I trained the model with hidden size of 32 and k=10. The results are quite similar to the previous k=10 experiment, but now only 9 features are dead.







It shows the reconstructions got worse although the amount of active features didn't drop drastically (23 active vs 30 active previously). Going even further down e.g. latent space = 15, the number of dead features seems to go to zero, but the reconstructions are visibly worse.

Going further

I think the next steps would be:

- Tackle dead features - learn the usage of auxiliary loss, initialization of encoder and decoder from the paper or other methods.
- Try different datasets and modality - e.g. faces, text embeddings.

https://cdn.openai.com/papers/sparse-autoencoders.pdf https://openai.com/index/extracting-concepts-from-gpt-4/